

Advanced Lesson 1 - Exercises: The Mathematica Interface

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MATH 7013 - Advanced Mathematical Modeling

1: Front-end tour - toolbar, magnification, Documentation Center

This exercise is entirely a walk through the notebook front end: show the toolbar, raise the magnification, open the Documentation Center, look up Integrate, and read the seven linked tutorials (Your First Mathematica Calculations, Arithmetic, Defining Variables, Defining Functions, Some Mathematical Functions, Symbolic Calculations, Numerical Calculations). The professor's own instruction is "Don't worry about an answer--just do it!" There is nothing to evaluate and nothing to submit.

One thing worth doing while the Documentation Center is open: the Basic Math Input palette (Palettes menu) supplies the typeset forms this lesson keeps asking for - the superscript template, the partial-derivative template, the integral-with-differential template, and the Pi, E and I characters. Every one of those has a keyboard equivalent, and the exercises below are written in ordinary linear syntax, which evaluates identically.

2: The numerical function N

$$N\left[\frac{1}{13}\right]$$

$$N\left[\frac{1}{13}, 50\right]$$

3: Assignment, and what "?" reports

```
Clear[a, A];
```

```
a A
```

```
?a
```

```
?A
```

```
a = 5
```

```
a A
```

```
?a
```

```
?A
```

```
A = 6
```

```
a A
```

```
?a
```

```
?A
```

```
A = 7
```

```
a A
```

```
?a
```

```
?A
```

4: Sin versus sin

```
?Sin
```

```
?sin
```

5: Symbols as units

```
Clear[a, A, feet, sec];
```

```
a
```

```
A
```

```
?a
```

```
?A
```

```
A =  $\frac{5 \text{ feet}}{\text{sec}}$ 
```

```
a
```

```
A
```

```
a = 10 feet
```

```
 $\frac{a}{A}$ 
```

```
?feet
```

```
?sec
```

6: Defining a function, differentiating, integrating, solving, plotting

```
Clear[x, f];
```

```
f[x_] = Sin[x];
```

```
{f[ $\pi$ ], f[ $\frac{\pi}{2}$ ], f[6.`]}
```

```
?f
```

```
 $\partial_x f[x]$ 
```

```
f'[x]
```

```
 $\int f[x] \, dx$ 
```

```
Solve[3 f[x] + 4 == 0, x]
```

```
Plot[f[x], {x, 0, 2  $\pi$ }]
```

7: $f[x] = x^2$ versus $f[x_] = E^x$ versus $f[2] = \text{Pi}$

```
Clear[a, x, f, y];
```

```
f[x] =  $x^2$ ;
```

```
f[x_] =  $e^x$ ;
```

```
f[2] =  $\pi$ ;
```

```
{f[x], f[a], f[y], f[2], f[3]}
```

```
?f
```

8: Immediate versus delayed assignment with RandomReal

```
Clear[f1, f2, x];
```

```
?RandomReal
```

```

f1[x_] = RandomReal[] x;
{f1[x], f1[x], f1[x]}

Plot[f1[x], {x, 0, 1}]

?f1

f2[x_] := RandomReal[] x;
{f2[x], f2[x], f2[x]}

Plot[f2[x], {x, 0, 1}]

?f2

```

9: Derivative, integral, radical equation, and a cubic plot

```

Clear[x, n];
 $\partial_x x^n$ 
 $\int x^n dx$ 
Solve[x +  $\sqrt{x}$  == 9, x]
Plot[x3 - 2 x + 1, {x, -2, 2}]

```

10: Wildcard searches on the name Style

```

?Style

Names["Style*"]

Length[Names["Style*"]]

Names["*Style"]

Length[Names["*Style"]]

Names["*Style*"]

Length[Names["*Style*"]]

```